

$$e^{i\phi} = \cos \phi + i \sin \phi$$

$\phi$  ANGLE IN  
RADIANS

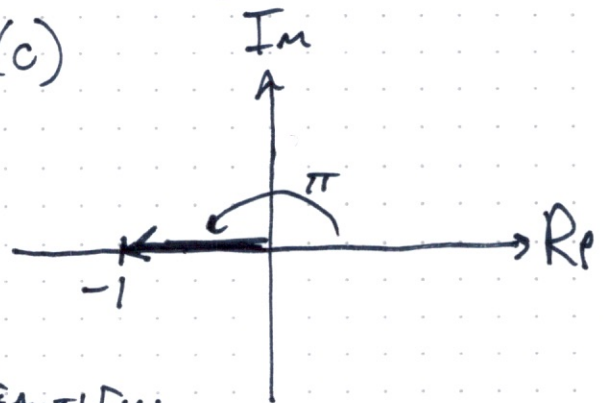
$$\phi = \pi$$

$$\phi = \pi/2$$

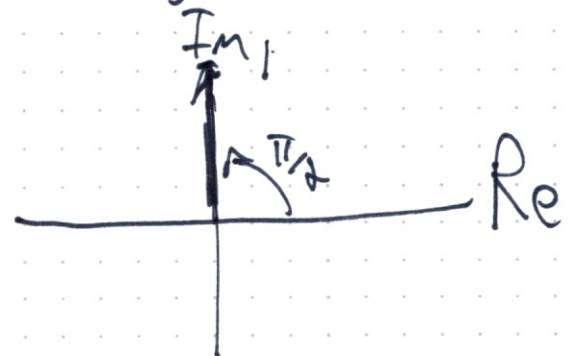
$$\begin{aligned} e^{i\pi} &= \cos(\pi) + i \sin(\pi) \\ &= -1 + i(0) \\ &= -1 \end{aligned}$$

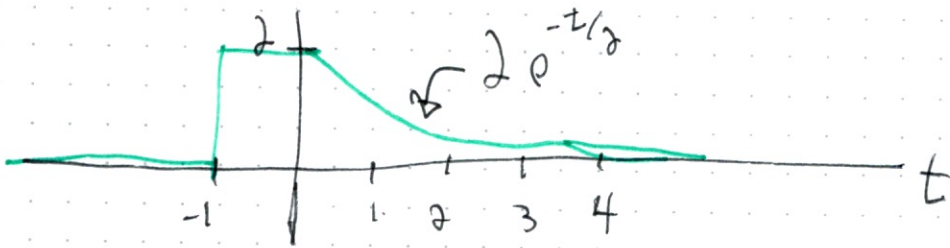
$$\underline{e^{i\pi} + 1 = 0}$$

← BEAUTIFUL.



$$\begin{aligned} e^{i\pi/2} &= \cos(\pi/2) + i \sin(\pi/2) \\ &= 0 + i(1) \\ &= i \end{aligned}$$





$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

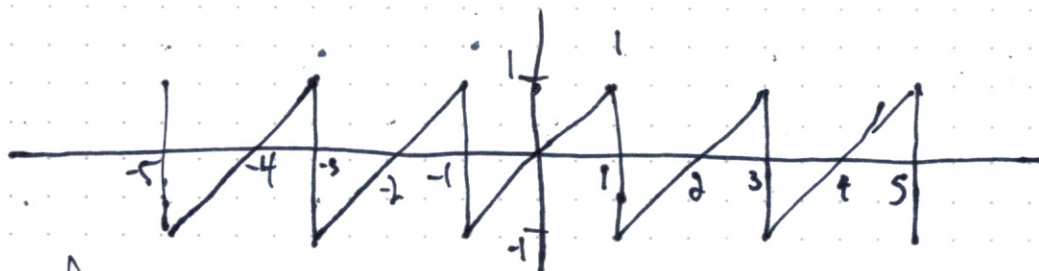
$$= \int_{-\infty}^{-1} 0 dt + \int_{-1}^0 |2|^2 dt + \int_0^{\infty} |2e^{-t/2}|^2 dt$$

$$= \int_{-1}^0 4 dt + \int_0^{\infty} 4e^{-t} dt$$

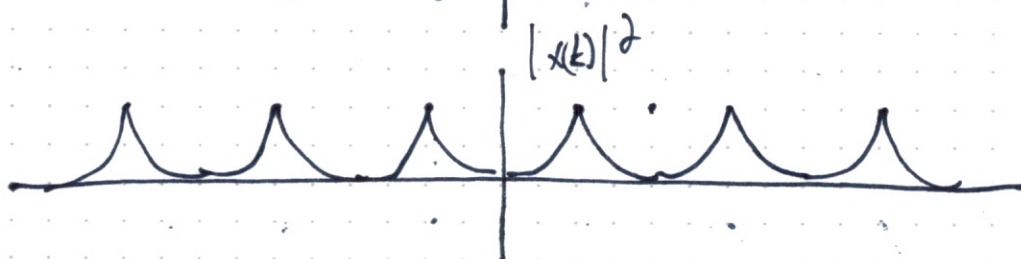
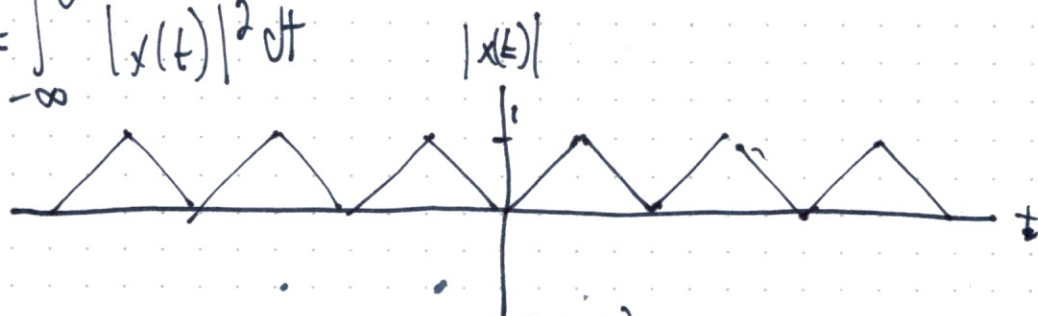
$$= 4t \Big|_{-1}^0 + 4(-e^{-t} \Big|_0^{\infty})$$

$$= \cancel{4(0)} - (4)(-1) + \cancel{4(-e^0)} - 4(-e^0)$$

$$= 4 + 4 = 8$$



$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt$$



$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt \rightarrow \infty$$

INSTEAD USE POWER

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt$$

IN THIS CASE  $x(t)$  IS PERIODIC,  
SO WE CAN CHANGE THE LIMITS  
OF INTEGRATION!

ONLY NEED TO SOLVE  $P_x$  OVER ONE PERIOD, FROM  $t = -1 \rightarrow t = 1$

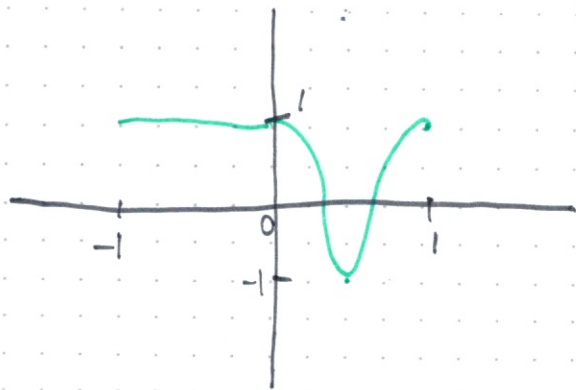
$T = 2$   
PERIOD

$$P_x = \frac{1}{2} \int_{-1}^1 |x(t)|^2 dt = \frac{1}{2} \int_{-1}^1 t^2 dt$$

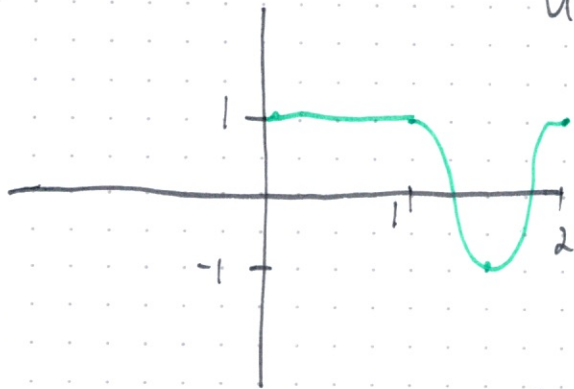
$$= \frac{1}{2} \left( \frac{1}{3} t^3 \Big|_{-1}^1 \right) = \frac{1}{2} \left( \frac{1}{3} (1)^3 - \frac{1}{3} (-1)^3 \right) = \frac{1}{2} \left( \frac{1}{3} + \frac{1}{3} \right)$$

$$P_x = \frac{1}{3}$$

$$u(t)$$



$$u(t) \rightarrow u(t-1)$$



$$u(t-1) \rightarrow u(\frac{1}{2}t-1)$$

